

Group Forecast and Furious
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Problem Statement

Financial markets are complex and dynamic systems in which asset prices evolve under the influence of both quantitative factors, such as historical price movements, and qualitative factors, such as the arrival of new information and investor sentiments. Traditional approaches to stock market prediction have primarily relied on historical price data and technical indicators; however, these methods often struggle to consistently capture short-term market movements due to the non-linear, volatile nature of financial time series data. Furthermore, the efficient market hypothesis suggests that asset prices rapidly incorporate publicly available information, making precise price forecasting inherently challenging (Fama, 1970).

Research has shown that financial news and media content can influence investor behavior and market outcomes by shaping sentiment and expectations. A prominent example of this would include press releases from President Donald Trump and these announcements can result in volatile movements within the stock market. Studies indicate that textual information from news sources contains predictive signals related to future stock returns, particularly when analyzed using natural language processing (NLP) techniques tailored to financial contexts (Tetlock, 2007; Loughran & McDonald, 2011). Rather than predicting exact future prices, several studies have demonstrated that modeling the direction of returns (i.e., whether prices increase or decrease) provides a more robust and practical objective for short-horizon market prediction tasks (Krauss et al., 2017).

Despite these advancements, it remains unclear whether integrating news-based sentiment features with historical price and technical indicators can consistently improve next-period return direction predictions compared to models that rely solely on price data (classical forecasting prediction models). This project aims to address this gap by developing and evaluating a predictive analytics framework that combines structured historical market data with unstructured financial news data processed through NLP techniques. By focusing on next-period return direction classification rather than exact price prediction, the study seeks to assess whether hybrid, multimodal models can enhance predictive performance while remaining aligned with established financial theory and realistic market constraints (Bollen et al., 2011).

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Articulation of Value

A model that accurately forecasts the price movement of a stock given its price history and sentiment analysis has a large potential monetary value. If an individual is confident that the value of a stock will decrease over the near term, they can sell this stock and invest in a more promising option. Conversely, if the individual is confident that the value of a stock will rise over the near term, they can buy low and receive a profit when they ultimately decide to sell. Each stock can be evaluated individually, and then summed up across a portfolio to receive the total monetary gain.

A secondary value that will derive from an accurate price prediction model is the mitigation of risk. A risk-seeking stockholder may place large investments on particular accounts, either winning or losing big depending on the trajectory of the stock. The reward for a correct decision here is enticing, but the consequence for an incorrect decision is potentially ruinous for a financial account. However, a model that is more confident of success than the current stock trader can balance these risks and minimize the chance of a loss.

The last value to mention is not related to a correctly forecasted change in price, but rather understanding consumer sentiment about a particular stock. This is particularly helpful for a publicly traded company, as their valuation is tied to a large group of users, many of whom post their sentiments on social media or news outlets. Understanding the sentiment of their current (and potential) stakeholders can help company leaders determine the effects of their past decisions and help influence future ones if they understand whether consumers are happy or dissatisfied with the current state of affairs.

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Calculation of Potential Economic Value

For the purpose of this section, we will focus primarily on the monetary economic value that an accurate price-predicting model can bring. As discussed earlier, this model can bring additional value by mitigating risk and increasing awareness of consumer sentiment on the stock.

To calculate the monetary gain of a correct model, some basic assumptions need to be created. Some of these are extremely presumptive, but are necessary for a valid calculation:

- The initial stock investment is \$1 MM
- Once a day, the stock trader will decide whether to buy more of the stock or sell
- A correct decision will increase the value of the investment 1% the following day while an incorrect decision will decrease the value of the investment 1% the following day.
- The stockholder can guarantee an 3.5% annual return (the current yield of a twelve-month treasury bond according to [Bloomberg](#))
- A successful model will have a 52/48 chance of correctly predicting whether the stock price will increase or decrease the following day (slightly increased from a completely random 50/50 split)
- There are no transactional costs to buying or selling the stock

Next are the steps for the expected return on investment over the course of 1 year, or 252 trading days.

Current State:

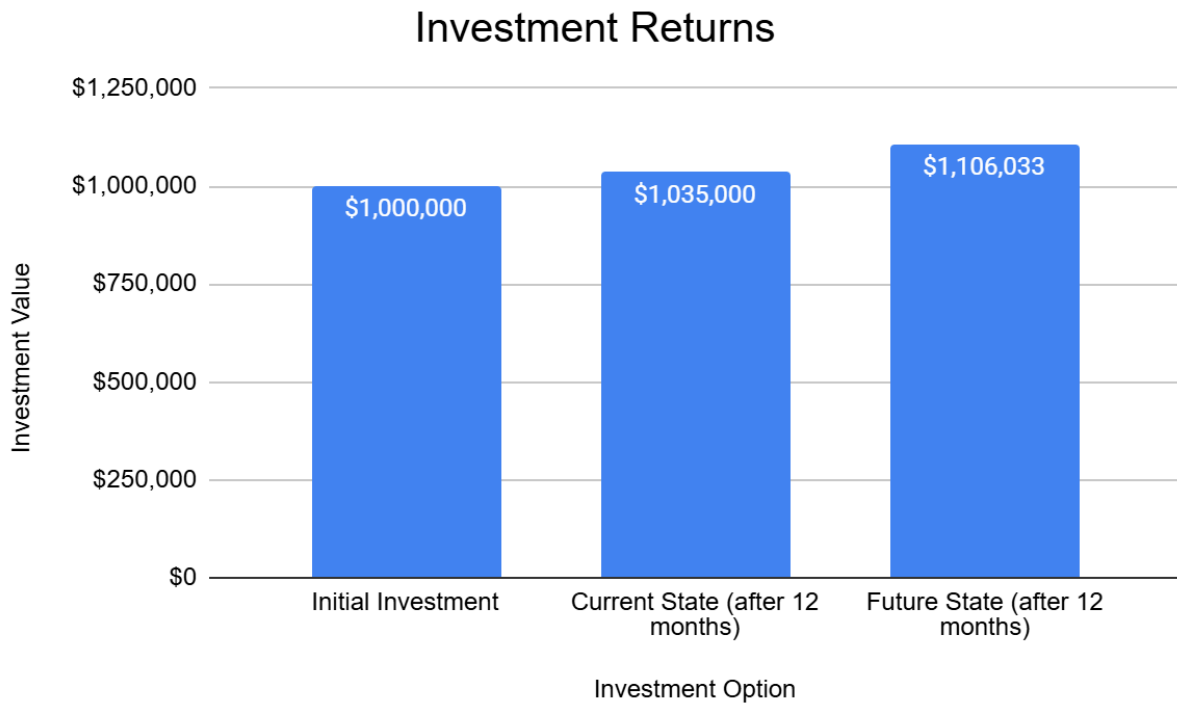
- The guaranteed return for a \$1MM investment is $\$1,000,000 \times 1.035 = \$1,035,000$.

Future State:

- The expected daily multiplier of a model with 52% accuracy is $.52(1.01) + .48(.99) = 1.0004$
- Expected Return = $1,000,000 \times (1.0004)^{252} = \$1,106,033$

Return on Investment:

- Current State - Future State = $\$1,106,033 - \$1,035,000 = \$71,033$ (7.1% of original investment)



The estimated value increase of a 52% accurate model over the current state guarantee of a 3.5% annual return is approximately \$71,033 over the next twelve months. This value would increase with a more accurate model, or if the guaranteed return were to decrease from 3.5%.

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Project Plan

The following is a project plan that outlines the specific steps needed to complete the project in 13 weeks:

- Week 1: Identify the problem statement and dataset
 - We will first identify a problem that can be solved with modeling. We will also identify a public dataset that we can use to solve this problem. Then, we will calculate the potential economic value of solving the problem. Finally, we will create a project plan that outlines the steps we will take to complete the project.
- Week 2: Explore the dataset
 - We will ingest the dataset into our Python environment and begin to explore its features. We will then identify which feature should be the prediction target and which features should serve as the predictors.
- Week 3: Perform exploratory data analysis (EDA)
 - In this step, we will split the dataset into training, validation, and test sets to prepare for modeling. We will also further examine the data by looking at summary statistics, correlations between features, number of missing values, etc.
- Week 4: Data processing
 - During this week, we use what we learned during EDA to clean and process our data. This may involve imputing missing values, standardizing or scaling features, and removing features. It will be important to consider potential data leakage in this step, so processing should be done to the train, validation, and test sets separately.
- Week 5: Feature engineering
 - Here, we will potentially augment our data if necessary. For example, our stock data may need a feature added to control for the number of trading days in each month.
- Weeks 6-8: Build models
 - We will then build three unique models to make predictions over the course of three weeks. We will train the models on the training dataset. It will be important to focus on tuning each model's hyperparameters for optimization.
- Week 9: Select the best model
 - In this step, we will evaluate each model's performance on the validation data. We will select which performance metric is the best for judging the performance of the models in our particular case and select the model that performs best based on that metric. We will then evaluate how the best model performs on the unseen test data.
- Week 10: Data-centric AI approach

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- For this week, we will look at a data-centric AI approach to improving our models by focusing on improving the quality of our data.
- Week 11: Risk, bias, and ethical analysis
 - Recognizing the potential ethical challenges associated with our model is essential. In this stage, we will examine these concerns and outline steps to address them.
- Week 12: Deployment
 - We will then recreate the process of deploying a model and monitor the deployment for potential problems.
- Week 13: Finalize project
 - Using all of the work from the past 12 weeks, we will finalize the project by putting everything together. We will also present our results.

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Dataset

Our main dataset will be time series stock data for Apple Inc (NASDAQ:AAPL). Apple has been a massive technology company that has been producing widely popular hardware and software for decades. Not only is there adequate stock data available for Apple, but there is also plenty of news about the company daily. Our dataset is sourced from Capital IQ and contains daily closing share prices for AAPL stock over the last 5 years (January, 2005 to January, 2026). By gathering daily share price, we can potentially calculate monthly or quarterly share price, depending on which direction we want to take our analysis. If we choose to use monthly or quarterly data, we can also further augment the dataset to include the number of trading days in each month or quarter. While the stock data is time series, we will not be trying to forecast the exact values of future stock price, but we will instead be using it to predict the direction of next-period returns.

We will also be using the Apple Stock (AAPL): Historical Financial News Data dataset from Kaggle. This dataset contains detailed information from news articles related to AAPL stock. There are 29,752 entries ranging from February 19, 2016 to November 27, 2024. Each entry represents a singular article. The dataset contains sentiment analysis information that quantifies the tone of each article. The exact sentiment analysis features include 'sentiment polarity', which tells us the overall sentiment of the article, 'sentiment negative', which measures the proportion of negative sentiment in the article, 'sentiment neutral', which measure the proportion of neutral sentiment, and 'sentiment positive' which measures the proportion of positive sentiment. For this project, we will rely on sentiment polarity, which can be a value ranging from -1 (negative) to 1 (positive). We aim to use the average sentiment polarity of all Apple news in a particular period to predict the direction AAPL stock will move in a future period.

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Modeling

To address the problem of predicting next-period return direction, the most appropriate approach is supervised learning. Supervised learning is suitable because the model is trained on historical data where the outcome is already known. Each observation includes input features such as past price movements, technical indicators (features), and news-based sentiment measures, along with a labeled outcome indicating whether the stock's return in the next time period was positive or negative. Having this labeled target variable allows the model to learn patterns that link the inputs to the observed outcome. More specifically, this project represents a classification problem rather than a regression problem. Although stock returns are numeric values, the goal of the project is not to predict the exact size of the return. Instead, the objective is to determine the direction of the return in the next period. By converting returns into categories such as "up" or "down," the modeling task becomes one of assigning observations to predefined classes, which is the defining characteristic of classification.

Within classification, this problem is best described as binary classification. The target variable has two possible outcomes: a positive return or a negative return in the next period. This binary setup simplifies the modeling task, makes evaluation more straightforward, and aligns well with common financial decision-making scenarios, such as whether to buy or hold an asset. Binary classification models also allow for the use of evaluation metrics like accuracy, precision, recall, and AUC, which are well-suited for assessing directional predictions. Overall, using supervised binary classification provides a clear and practical framework for solving the problem. It allows the model to learn from historical examples, incorporates both numerical and textual data, and focuses on a realistic prediction goal that avoids the challenges associated with forecasting exact stock prices.

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